Universal Bridge Attestation Soundness

B-Chain ∘ M-Chain, and the M/T/F separation as a security boundary

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Abstract

The Lux universal bridge honours a cross-chain message only when an M-of-N *threshold attestation*, produced by the M-Chain over a message that is *final on the source chain*, is presented on the destination chain. We formalise this composition and prove *bridge attestation soundness*: forging acceptance of a message that was never source-finalised reduces to either a threshold-signature forgery or a source-chain finality violation. Both are negligible — the first by the EUF-CMA security of the M-Chain signing scheme (CGGMP21 / FROST), the second by the QuasarCert *No double-finalization* theorem (`quasar-cert-soundness.tex` §"Finality Admission"). We then show that this reduction *depends on* the M/T/F separation of concerns (LP-7050, LP-7100, LP-8200): braiding the M-Chain signing shares with F-Chain FHE compute in one VM shares a fault domain between the bridge-custody key and the encrypted-compute path, silently weakening the unforgeability assumption the whole reduction rests on. Separation is therefore a security boundary, not an aesthetic one.

1 Chains and concerns

Definition 1 (Concern-owning chains)**.** *Per LP-134 / LP-7050 the roster is $P, C, X, Q, Z, A, B, M, F$, and each of the former T-Chain concerns has exactly one owner:*

M-Chain (MVM, LP-7100): *runs M-of-N MPC signing ceremonies (CGGMP21 threshold ECDSA, FROST threshold Schnorr, Pulsar-general) for bridge custody of external-chain wallets. It* attests*; it does not decide finality and does not compute on ciphertexts.*

F-Chain (FVM, LP-8200): *runs FHE (TFHE bootstrap-key DKG, encrypted-EVM execution, no-reconstruct threshold decryption). It* computes on encrypted data*; it holds no bridge-custody signing key.*

B-Chain (`bridgevm`, LP-6000): *transports and relays cross-chain messages (teleport). It* carries *attestations; it mints no signature and finalises no source block.*

Source chain S (any Lux L1/L2): *produces QuasarCert finality over its own blocks (§2).*

The M-Chain and F-Chain share *the* ***`thresholdvm`*** *substrate library but are distinct chains; §5 shows why the distinctness is load-bearing.*

2 Source-chain finality (the input)

Definition 2 (Source finality)**.** *For an event slot e on source chain S (a height, or a height-and-log-index), let* $\mathrm{Final}_S(e, m) = 1$ *iff message m is committed at e by a verified QuasarCert over the canonical execution commitment at e's height.*

Lemma 3 (Finality soundness of S)**.** *Below $S/3$ Byzantine stake on S, for every event slot e there is at most one m with* $\mathrm{Final}_S(e, m) = 1$.

Proof. Immediate from the QuasarCert *No double-finalization* theorem (`quasar-cert-soundness.tex` §"Finality Admission"): two certificates at one height over different canonical commitments require an honest validator to have signed both, which the durable decided-height sign gate forbids. A single canonical commitment fixes a single m at e. □

3 The M-Chain attestation and the B-Chain relay

Definition 4 (Threshold attestation scheme)**.** *The M-Chain committee of size N holds an M-of-N threshold key with aggregate public key* apk *(established by a DKG; no party holds the secret).* $\mathsf{TS.Sign}$ *is the interactive ceremony requiring $\geq M$ honest participants;* $\mathsf{TS.Verify}(\mathsf{apk}, x, \sigma) \to \{0, 1\}$. TS *is EUF-CMA: an adversary controlling $< M$ committee members and making chosen-message ceremony queries forges a signature on a fresh x with advantage at most ϵ_{TS} (CGGMP21 under enhanced-existential unforgeability of ECDSA; FROST under the OMDL / AGM analyses).*

Definition 5 (Attested bridge message and acceptance)**.** *The M-Chain emits an attestation only after checking source finality:*

$$x = (\mathsf{srcID}, e, m, \mathsf{finalityProof}), \quad \sigma \leftarrow \mathsf{TS.Sign}(x) \text{ iff } \mathrm{Final}_S(e, m) = 1.$$

B-Chain relays (x, σ) to destination D. D ***accepts*** *m for e iff*

$$\mathsf{TS.Verify}(\mathsf{apk}, x, \sigma) = 1 \ \wedge \ x.m = m \ \wedge \ e \notin \mathsf{consumed}_D,$$

and then records $e \in \mathsf{consumed}_D$ (single-spend / replay guard).

4 Soundness

Theorem 6 (Bridge attestation soundness)**.** *Assume (i) fewer than M of the N M-Chain committee members are Byzantine, (ii) S has $< S/3$ Byzantine stake, and (iii)* TS *is EUF-CMA with advantage ϵ_{TS}. Then the probability that destination D accepts, for some event slot e, a message $m^\star$ with $\mathrm{Final}_S(e, m^\star) = 0$ is at most ϵ_{TS}. Moreover D accepts at most one m per e, and that m is the unique source-finalised message for e.*

Proof. Suppose D accepts $m^\star$ for e. By Definition 5 there is a $\sigma^\star$ with $\mathsf{TS.Verify}(\mathsf{apk}, x^\star, \sigma^\star) = 1$ and $x^\star.m = m^\star$. Consider the message $x^\star = (\mathsf{srcID}, e, m^\star, \cdot)$.

Case A: $x^\star$ was never signed by the honest ceremony. Then $\sigma^\star$ is a valid signature on a message the honest committee never produced. By assumption (i) the adversary controls $< M$ members, so producing $\sigma^\star$ is exactly an EUF-CMA forgery against TS; this occurs with probability $\leq \epsilon_{\mathsf{TS}}$.

Case B: $x^\star$ was signed by the honest ceremony. By Definition 5 the ceremony signs $x^\star$ only when $\mathrm{Final}_S(e, m^\star) = 1$. Hence $m^\star$ *is* source-finalised, contradicting the hypothesis $\mathrm{Final}_S(e, m^\star) = 0$.

So a wrongful acceptance implies Case A, of probability $\leq \epsilon_{\mathsf{TS}}$. For the uniqueness clause: any accepted m falls in Case B, so $\mathrm{Final}_S(e, m) = 1$; by Lemma 3 at most one such m exists, and the $\mathsf{consumed}_D$ guard admits e only once, so D accepts exactly that unique m and no replay. □ □

Remark 7 (Two disjoint trust hypotheses)**.** *Bridge soundness is the* conjunction *of two independent assumptions living on two different chains: threshold honesty on the M-Chain (hypothesis*

i) and stake honesty on the source chain (hypothesis ii). Neither implies the other, and each is verified in its own domain — exactly the "one concern, one chain" discipline of Definition 1. A single actor cannot forge a bridge acceptance without either compromising M M-Chain signers or breaking source finality.

5 Why separation of concerns is load-bearing

Theorem 6 rests on hypothesis (iii): TS is EUF-CMA. That, in turn, presumes the threshold *signing shares* are confined to the M-Chain's fault domain — an adversary that can read or fault-inject a share does not need M colluding members.

Definition 8 (Fault domain)**.** *A* fault domain *is a unit of code + memory + accelerator state within which a fault (memory-disclosure bug, side channel, malicious GPU kernel, shared-address-space corruption) can move secret material. Two subsystems in the same VM/process share a fault domain.*

Lemma 9 (Braiding enlarges the trusted surface)**.** *Let* TS*'s signing shares and an FHE evaluation engine reside in one VM (the overloaded* `thresholdvm`*: MPC signing +* `vms/thresholdvm/fhe` *in one address space). Then the unforgeability advantage against* TS *is bounded not by* ϵ_{TS} *alone but by* $\epsilon_{\mathsf{TS}} + \epsilon_{\text{FHE-leak}}$*, where* $\epsilon_{\text{FHE-leak}}$ *is the probability that a fault in the FHE path (a bug in an encrypted-EVM opcode, a malicious bootstrap kernel, a decryption side channel) discloses or corrupts a signing share.*

Proof. An adversary now has two ways to obtain a forgery: the M-collusion path (bounded by ϵ_{TS}) *or* exercising the FHE path to exfiltrate/fault a share and sign directly (bounded by $\epsilon_{\text{FHE-leak}}$, since both live in one fault domain by Definition 8). The advantage is the union bound $\epsilon_{\mathsf{TS}} + \epsilon_{\text{FHE-leak}}$. The FHE surface is large (a full encrypted-EVM opcode set plus GPU bootstrap kernels), so $\epsilon_{\text{FHE-leak}}$ is neither zero nor small a priori. □

Corollary 10 (M/F separation is a security boundary)**.** *Placing MPC signing (M-Chain, LP-7100) and FHE compute (F-Chain, LP-8200) in* distinct *chains with* disjoint *fault domains restores hypothesis (iii) to* ϵ_{TS} *alone, and thus restores Theorem 6 to its stated bound. The separation is therefore not cosmetic: it removes the* $\epsilon_{\text{FHE-leak}}$ *term from bridge soundness. The two chains may share the* `thresholdvm` *substrate* library *and even* `constants.ThresholdVMID` *(disambiguated by* `CreateChainTx.BlockchainName`*, LP-134) without sharing a fault domain, provided the running instances do not co-reside signing shares with FHE evaluation state.*

Remark 11 (Implementation status)**.** *As of 2026-07-02 the node has not yet performed the split: the genesis registry (*`node/genesis/builder/registry.go`*) lists a single overloaded* `T-Chain` *entry and no M/F rows, and the* `thresholdvm` *package still braids MPC signing with* `vms/thresholdvm/fhe`*. Corollary 10 is thus the* target *security posture; LP-7050 §8 tracks the migration. Until the split lands, bridge soundness carries the* $\epsilon_{\text{FHE-leak}}$ *term of Lemma 9 and should be treated accordingly (keep the FHE surface off any node that holds live M-Chain custody shares).*

6 Conclusion

The universal bridge is sound exactly when two disjoint honesty hypotheses hold on two different chains, and its soundness proof *requires* the M/T/F separation to keep the threshold-signature assumption clean. Separation of concerns here is a theorem hypothesis, not a style preference.